LAYERBUILDER

Version 1.57
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LayerBuilder is a Windows based GUI program used to setup .layer files. The .layer files will then be processed to generate the .geo, .mater and .doping files, which are needed for creating the mesh and running the simulation.

1 Overview of Layerbuilder

The Layerbuilder application by Crosslight Software Inc. allows one to interactively input a device structure. With this utility it is much easier to create or modify device structures.

Figure 1 shows some of Layerbuilder's key featuers:

1. User-friendly dialog for input of various device sizes, material type, composition grading, contact type and other properties.

2. Powerful device editing capabilities. You can cut, copy, move, drag and drop layers and columns.

3. Layers with different material are assigned different colours and textures for easy identification.

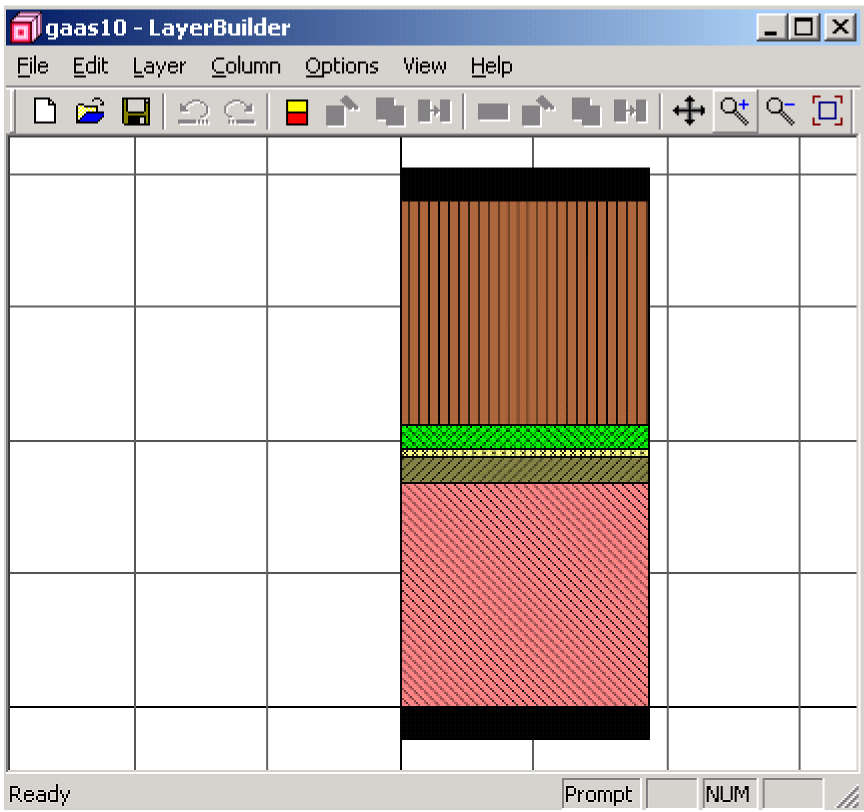


Figure 1: Overview of Layer builder

4. One key click to change or modify a layer.

2. A few concepts in LayerBuilder

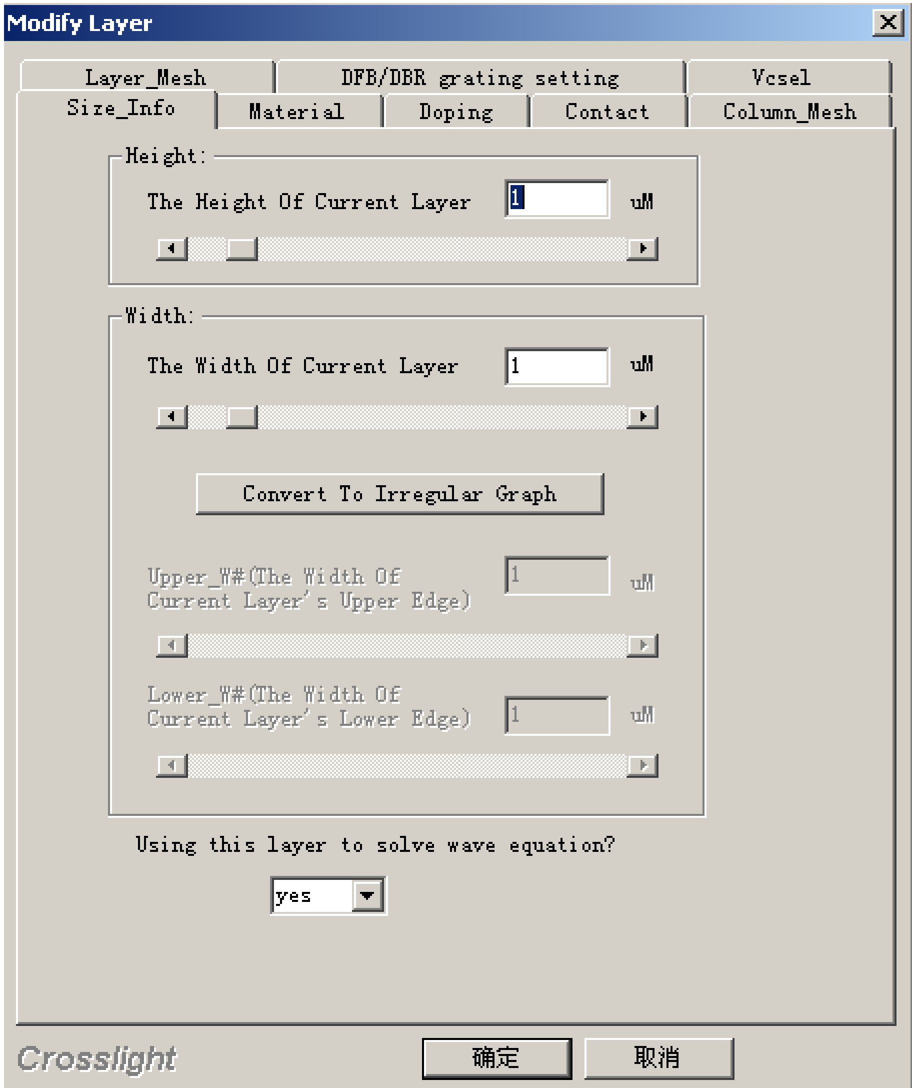
(2.1) LayerBuilder is mainly used to setup some devices with regular geometrical structures, and to set some physical properties which will be used by the main core simulator. A rectangular or taper form of a block of device is called a “layer”. When click the device with mouse, a block of device will be shaded, which corresponds to a layer.

(2.2) We call it a column if a group of geometrical layered structures stacked in vertical direction.

(2.3) The physical properties in LayerBuilder are corresponding to a layer.

(2.4) After a layer is chosen, its physical properties can be modified by clicking right button of the mouse (or through menu, tool bar). The physical properties can be modified through property dialog box.

(2.5) Size_Info properties dialog box



(2.5.1) Height

The height of the current layer. If this is modified, the height(s) of other layers belonging to adjacent column will be changed accordingly.

(2.5.2) Width

The height of the current layer. If it is modified, the width of the corresponding column will be changed accordingly.

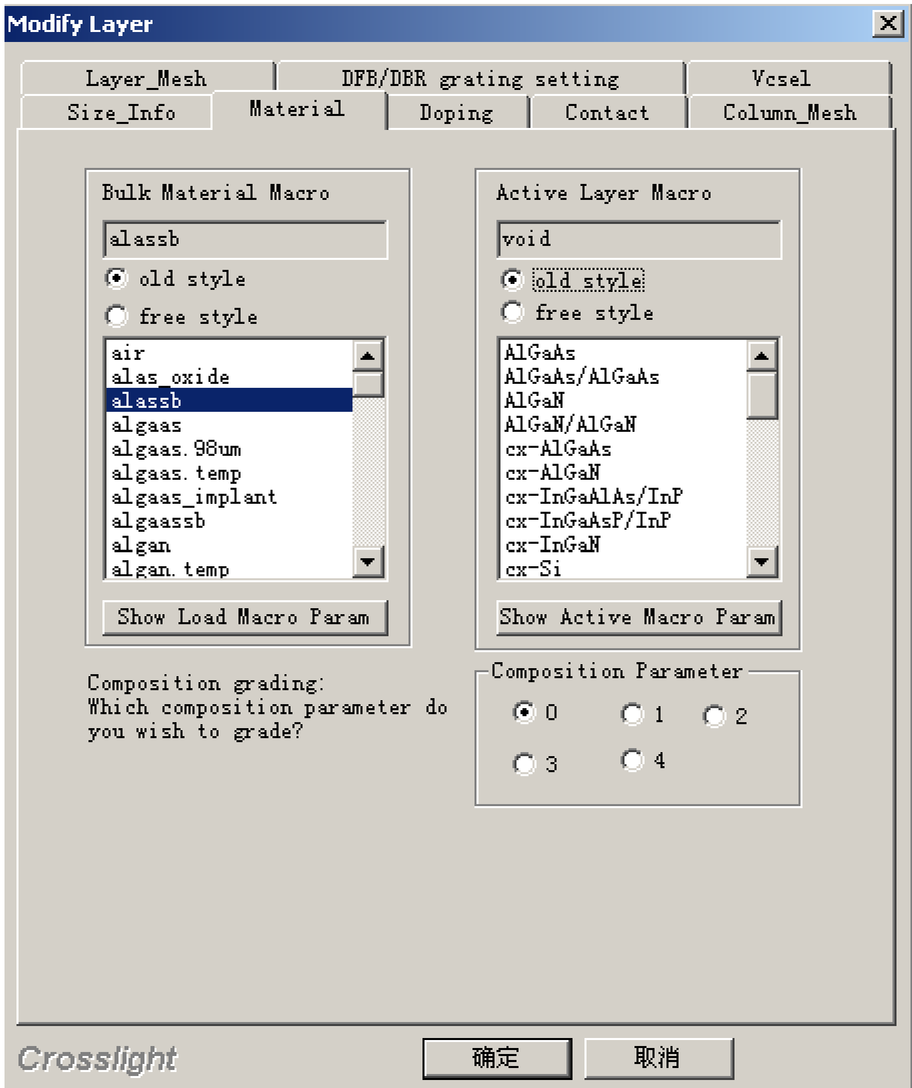
(2.5.3) Convert To Irregular Graph

Change the current layer from rectangular to trapezoid shape.

(2.5.4) Using this layer to solve wave equation?

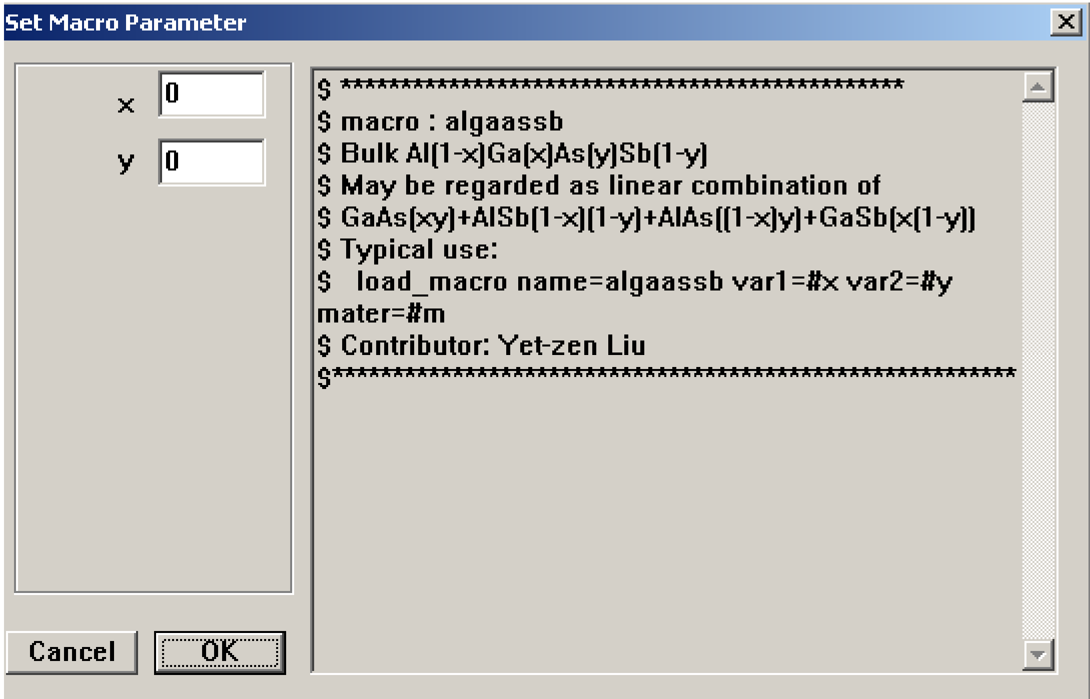
Choose if the current layer will be included in the solution of the wave equation. It may influence the simulation of the core simulator.

(2.6) Material property dialog box



(2.6.1) Bulk Material Macro

Set Bulk Material Macro parameters. In Bulk Material Macro List Box, a material name and double-click it, or push Show Load Macro Param button to set its parameter. Above the List Box, there are two buttons to choose old style or free style of material macro. Old style is here for the compatibility reason with older version of the simulator. For the difference between old style and free style, please read explanations in crosslight.mac and more.mac. Note: If the macro name is chosen with mouse in the list box, and the parameter is not set with the method mentioned above, the name of macro should be double clicked or push Show Load Macro button to make the selection effective. The dialog box of the parameter setting is as follows.

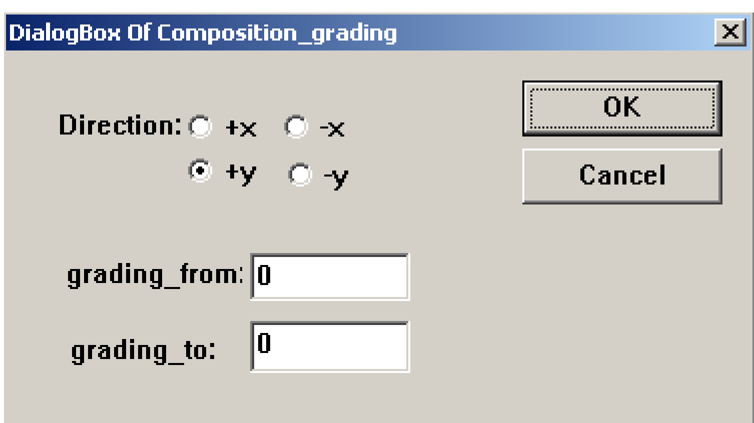


(2.6.2) Active Layer Macro

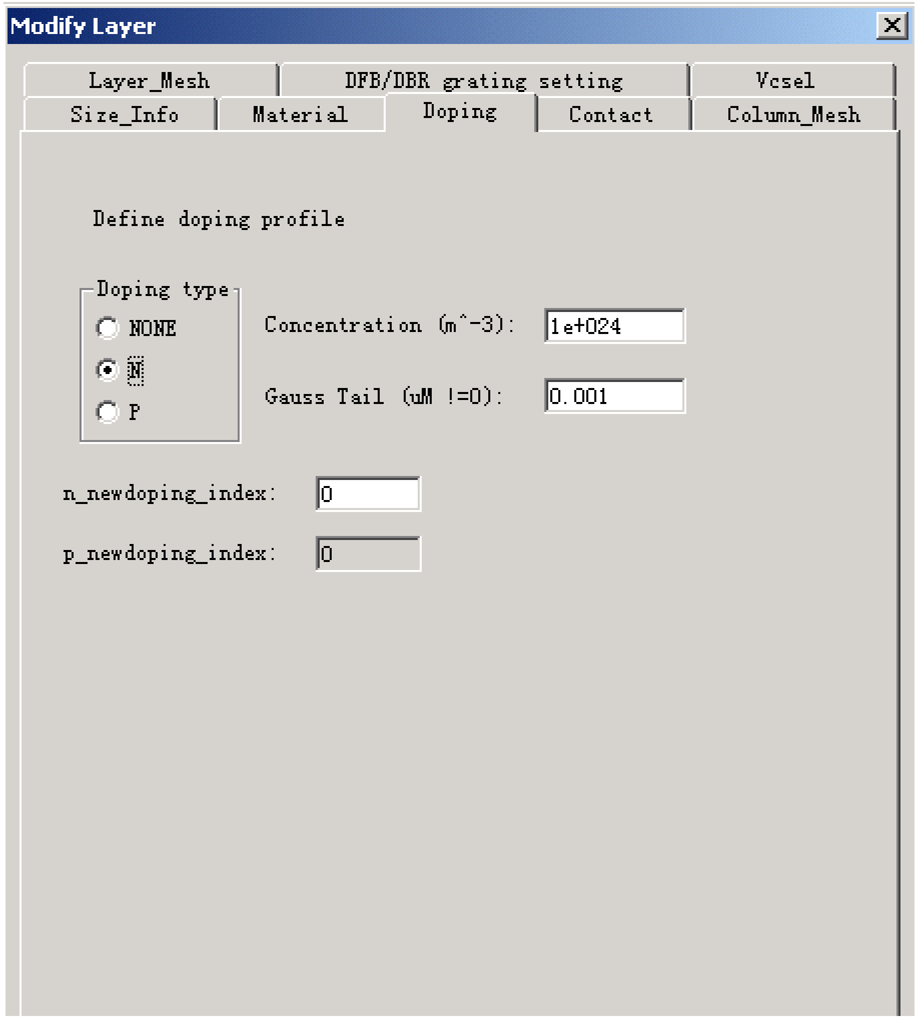
Set Active Layer Macro parameters. The same way as to change Bulk Material Macro.

(2.6.3) Comosition Parameter

Click 0, 1, 2, 3, 4 radio button to set parameters. Following is the dialog box for parameter modifications.

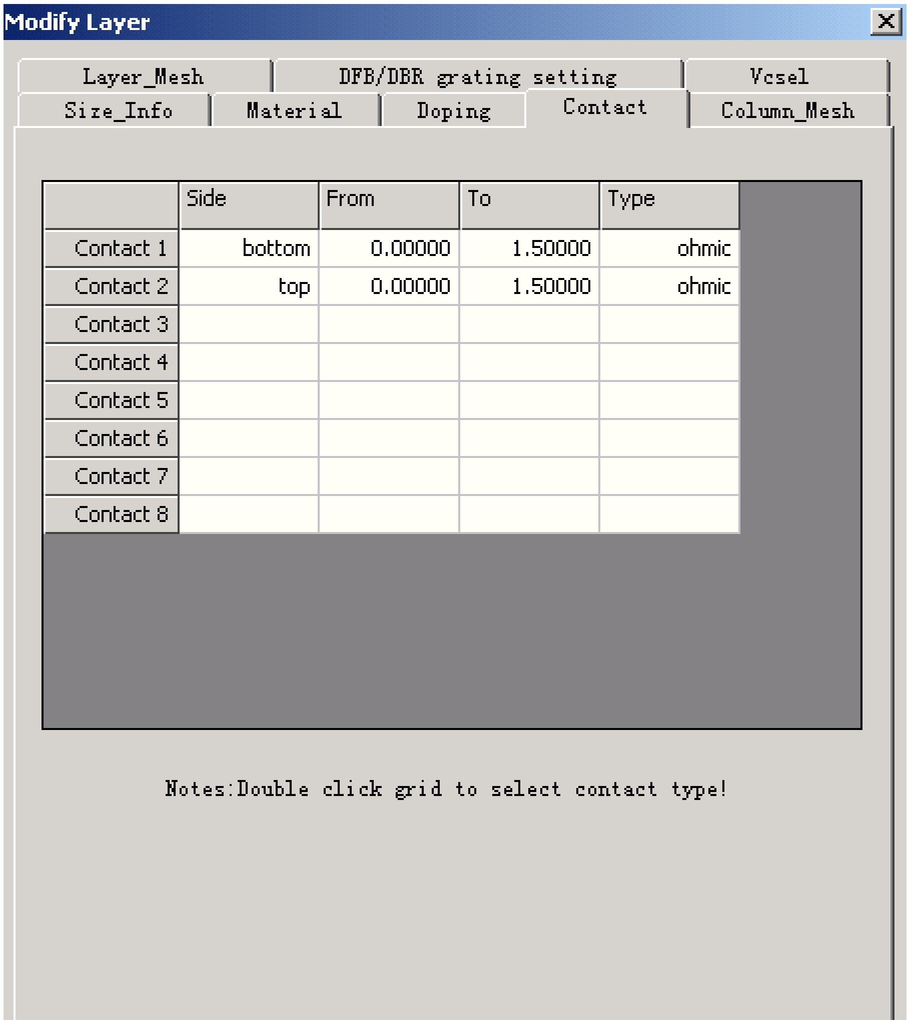


(2.7) Doping Properties



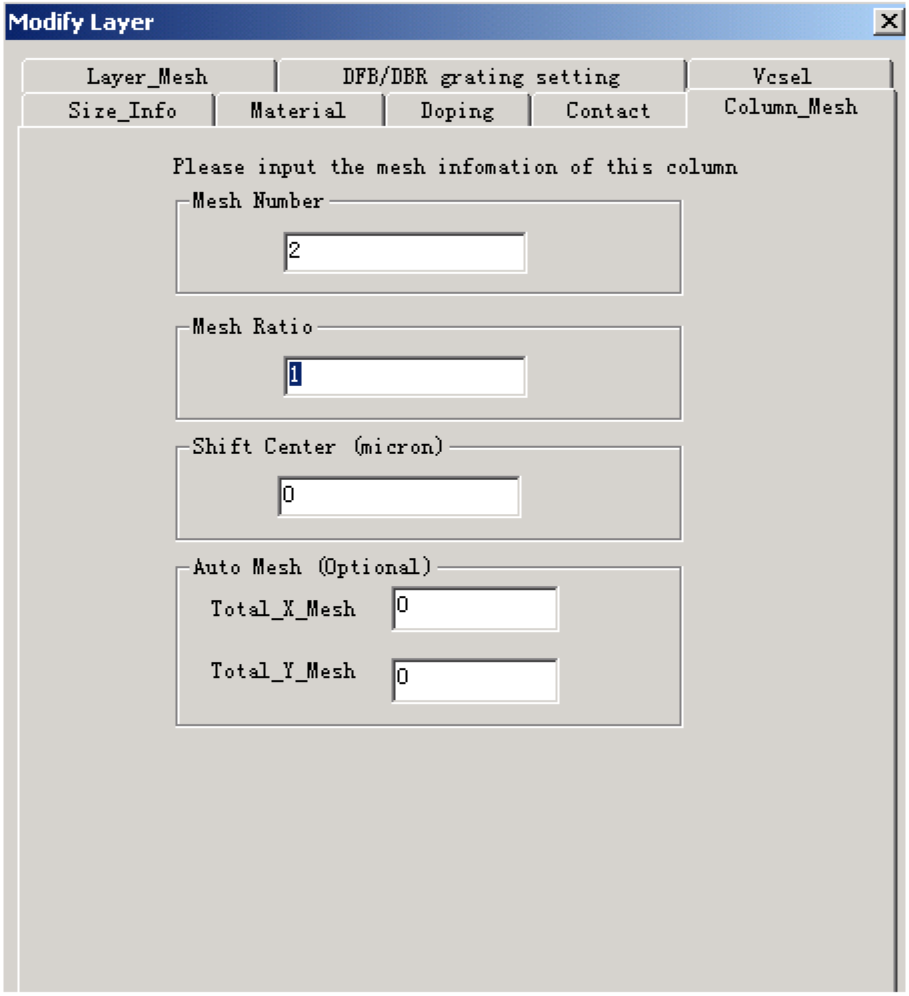
(2.7.1) Set Doping type and concentrations of the current layer.

(2.8) Contact properties



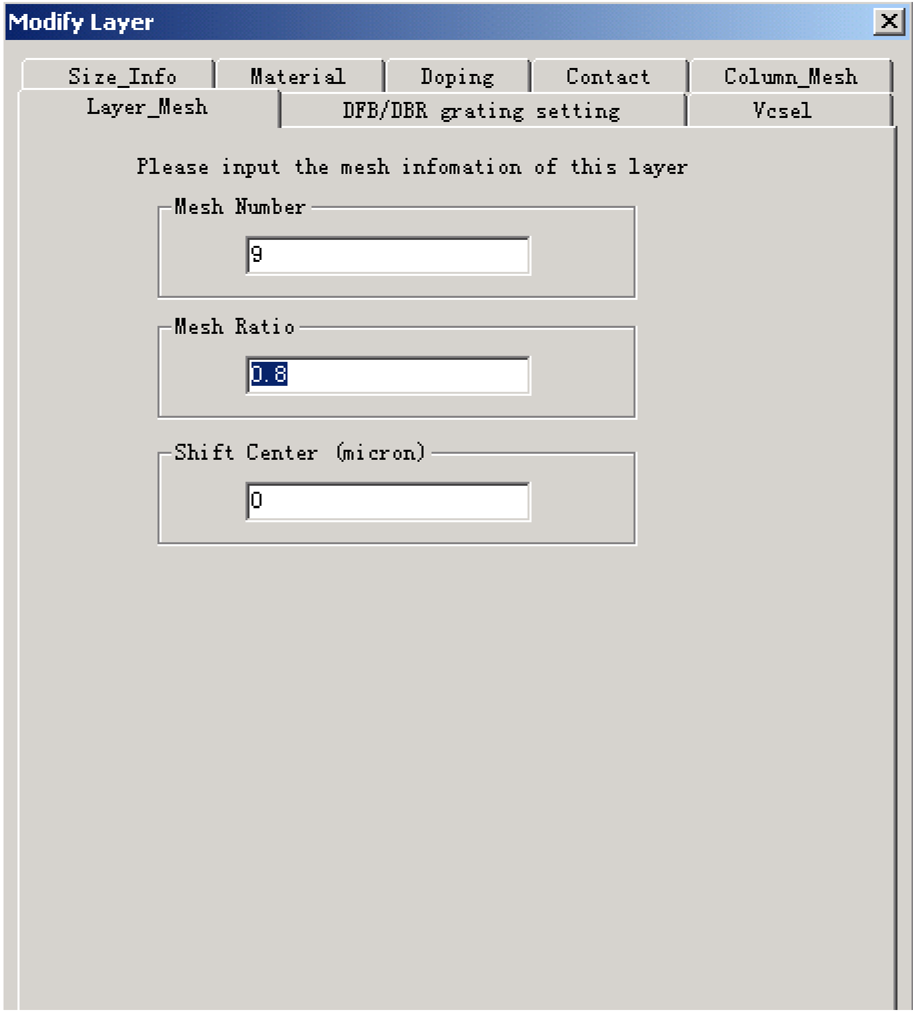
(2.8.1) Set Contact properties. For the Contact Side and Contact Type, user can double-click grid to select in the options.

(2.9) Column_Mesh Property



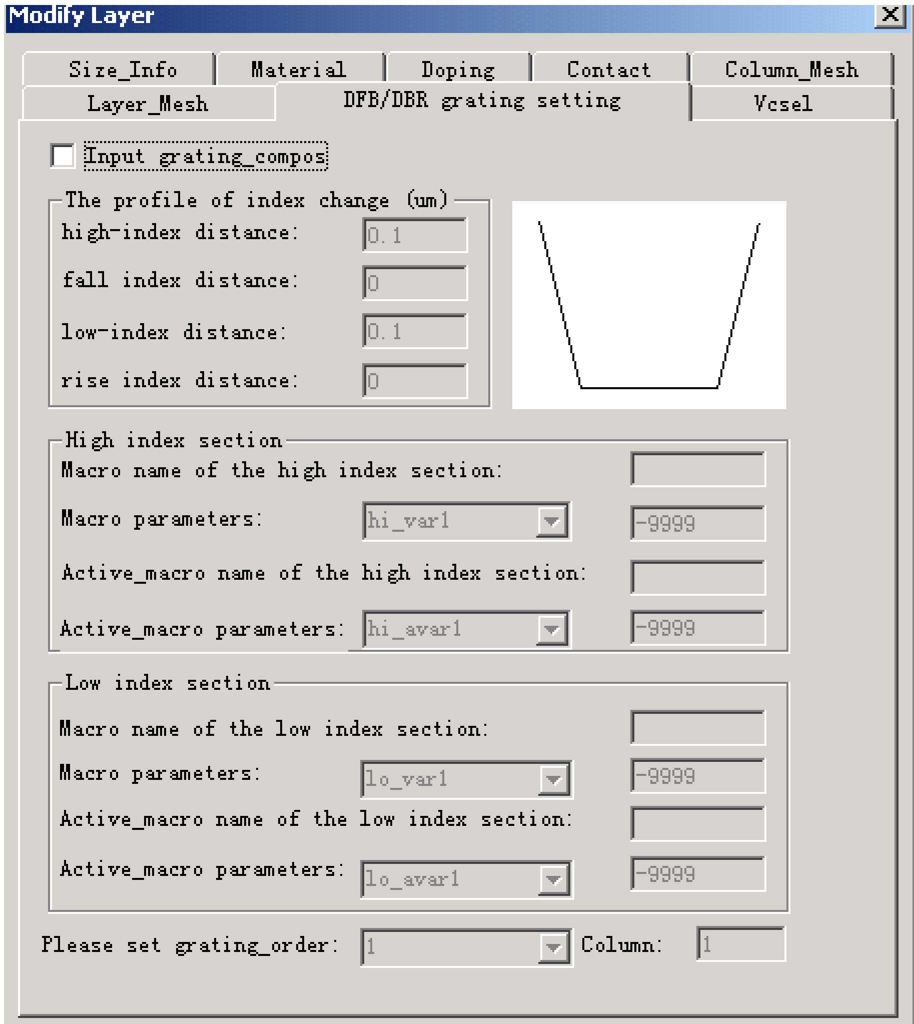
(2.9.1) Set mesh for a column that current layer belongs to.

(2.10) Layer_Mesh



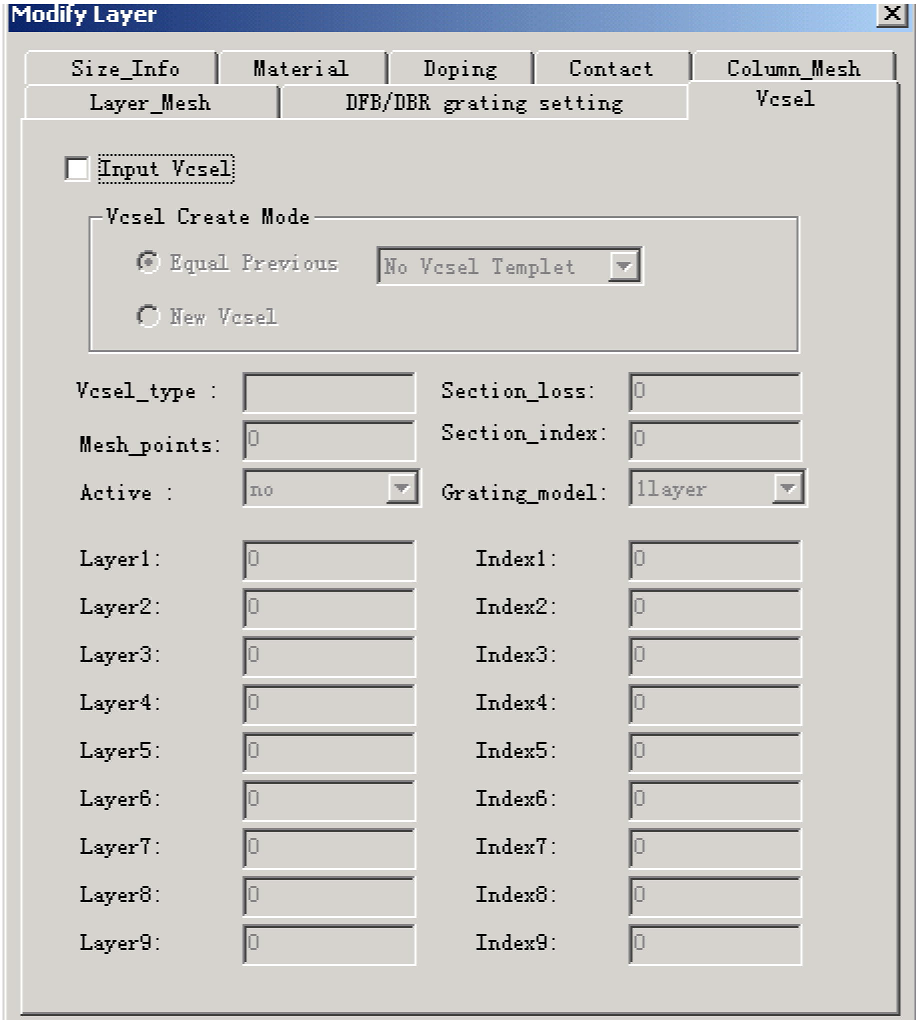
(2.10.1) Set mesh of current layer.

(2.11) DFB/DBR grating setting



(2.11.1) Set grating compos for the current layer. The user should check Input grating compos check box to make the setting effective.

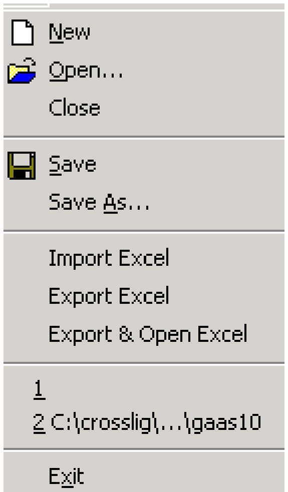
(2.12) Vcsel



(2.12.1) Set Vcsel related properties. To make these properties effective, the check box of Input Vcsel should be checked

3 LayerBuilder Menus

(3.1) File Menu



(3.1.1) New

Create a new layer. The current layer will be cleared.

(3.1.2) Open

Open a layer file.

(3.1.3) Close

Close current layer file.

(3.1.4) Save

Save current layer file.

(3.1.5) Save As

Save current layer file with a new name.

(3.1.6) Import Excel

Import Excel file. This Excel file contains detailed information about device structure.

User can open a layer file, then export to Excel format file under "Export Excel".

(3.1.7) Export Excel

Export current layer file with Excel format.

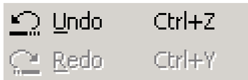
(3.1.8) Export & Open Excel

Export current layer file in Excel file format, and open with Microsoft Excel program.

(3.1.9) Exit

Exit LayerBuilder program.

(3.2) Edit Menu



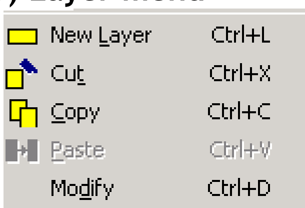
(3.2.1) Undo

Cancel previous operation.

(3.2.2) Redo

Redo previous operation.

(3.3) Layer menu



Layer menu is used to edit selected layer. To use layer, select a layer with mouse first. The selected layer is displayed with black colour.

(3.3.1) New Layer

Create a new layer above selected layer.

(3.3.2) Cut

Delete selected layer.

(3.3.3) Copy

Copy selected layer. User can paste it later.

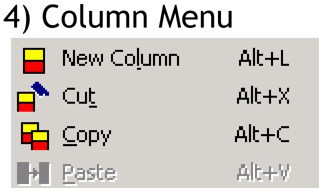
(3.3.4) Paste

Paste a layer, which is cut or copied beforehand, in the currently selected layer.

(3.3.5) Modify

Modify properties of the selected layer.

(3.4) Column Menu



Column menu is used to edit selected column. To use this menu, user should select the column by clicking a layer in that column.

(3.4.1) New Column

Create a new column next to the selected column.

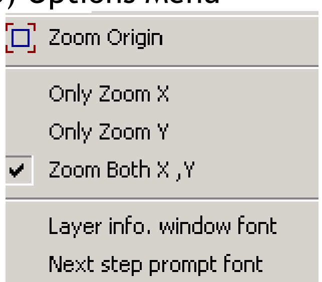
(3.4.2) Cut

Delete current column

(3.4.3) Copy
Copy current column

(3.4.4) Paste
Paste a column, which is cut or copied beforehand, next to the selected column.

(3.5) Options Menu



(3.5.1) Zoom Origin
Resume to original size and position.

(3.5.2) Only Zoom X
Zoom in x-direction

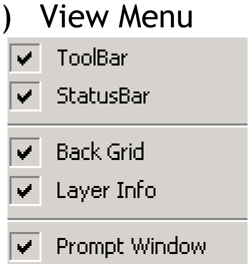
(3.5.3) Only Zoom Y
Zoom in y-direction

(3.5.4) Zoom Both X, Y
Zoom in x- and y-direction

(3.5.5) Layer info. window font
Set font of Layer info window

(3.5.6) Next step prompt font
Set font of the prompt window

(3.6) View Menu



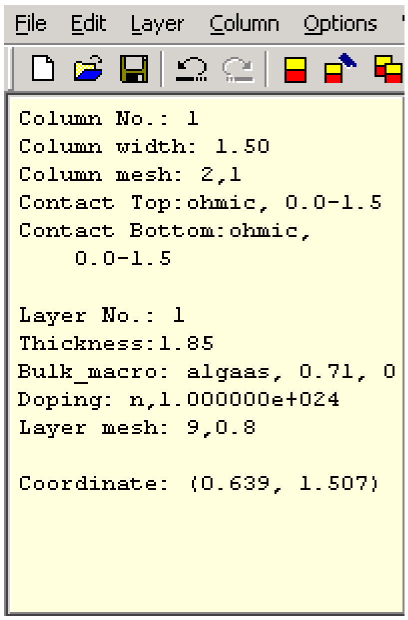
(3.6.1) ToolBar

On/Off tool bar

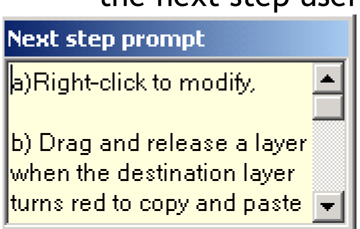
(3.6.2) StatusBar
On/Off status bar

(3.6.3) Back Grid
On/Off background grid line

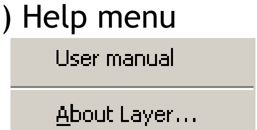
(3.6.4) Layer Info
On/off the layer info display window. To display layer information in the layer Info window, user should move the mouse to a layer. The Layer Info window looks like:



(3.6.5) Prompt Window
On/Off prompt window. Prompt window is used to show what the next step user might to do. The prompt window looks like:



(3.7) Help menu



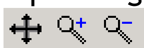
(3.7.1) User manual
Open LayerBuilder help manual

(3.7.2) About Layer…

Show LayerBuilder’s version information etc.

(4) Tool bar of LayerBuilder

The buttons on the Tool bar are used as short cut of the menu. The details of their usage are explained below. There are three buttons without corresponding menus. They are as follows:

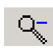


(4.1)  button

Use this button to move the whole layer view by dragging the screen with the shift key pressed.

(4.2)  button

This button is used to expand the view in all dimensions. User can use the corresponding menus to zoom x-, y-, or z-directions separately.

(4.3)  button

This button is used to shrink the view in all dimensions. User can use the corresponding menus to zoom x-, y-, or z-directions separately.

2 Drag and Drop Using the Mouse

An intuitive approach to edit/modify a device structure is to use the mouse to click on a piece of device material, drag it to a new location, and drop it there. This can be used to copy and move material pieces.

To copy a material piece, simply left click this piece and hold down the left mouse button while you move it to a new location and drop it there. The new location must be above another existing piece of material. You must wait until the existing piece changes colour indicating that your move will be accepted. Suppose you want to copy material A to the top of material B, you need to drag A and move it close to B. When B changes colour, drop it there and the new piece will appear above B.

To move a material piece to a new location, follow the above procedure, except you must also hold down the "Ctrl" key during the drag and drop.

We find it very convenient to create a device by copying from a few existing layers using the "drag-drop" techniques. Once we have copied new material pieces, we promptly modify their properties to suit our needs.

To a selected layer, user can click right button of mouse to pop up proper dialog box.

3 Tutorial Examples

In this tutorial section we show step by step how to create a device structure for Crosslight Software Simulators (LASTIP, PICS3D and APSYS) using the LayerBuilder GUI. We will take three examples. The first is a three-piece double heterostructure laser. The second is a simple 1D (1-column) laser diode single quantum well laser. The third is a 2D (2-column) QW ridge waveguide laser.

3.1 Example 1

A 1D 3-layer GaAs/AlGaAs laser diode structure with the following schematics:

Top Contact

- -

Al(0.5)Ga(0.5)As(p-doped)

1μm

- -

Al(0.1)Ga(0.9)As undoped(active)

0.2μm

- -

Al(0.5)Ga(0.5)As(n-doped)

1μm

- -

Bottom Contact

1. Set up the first layer of the only column of the device.
   In Layer builder a device is made of columns that contain several layers and in order to build a layer, the column should be set up at first.

   1) Two ways to start a new column:
      - Click the short-cut button for "New Column" on the toolbar.
      - From the "Column Edit" menu select "New Column."

   See
   Figure 2: Setup a New Column

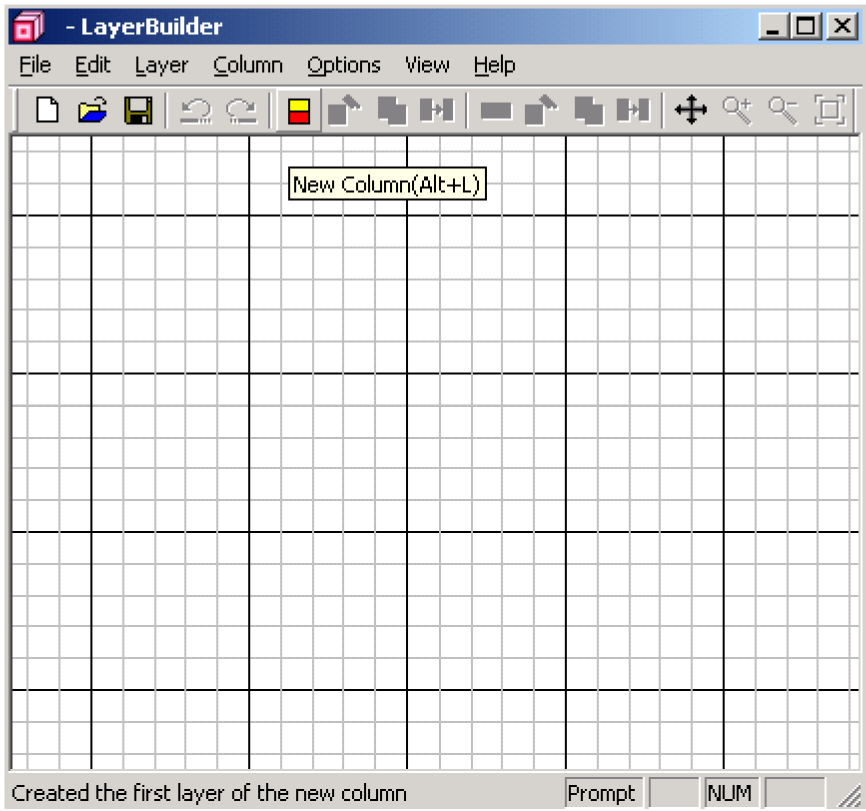


Figure 2: Setup a New Column

2) A properties dialogue box will pop up, where the data for the first layer of this column should be input.
   Refer to Figure 3.

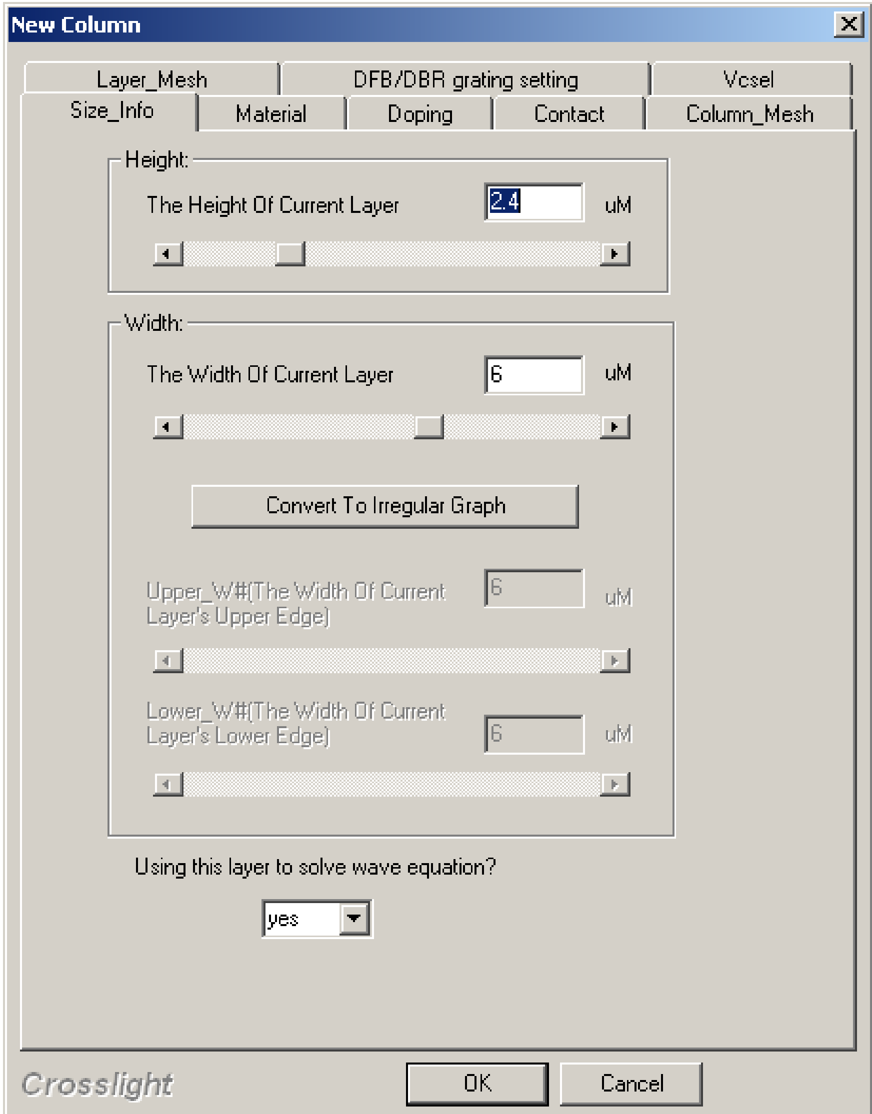
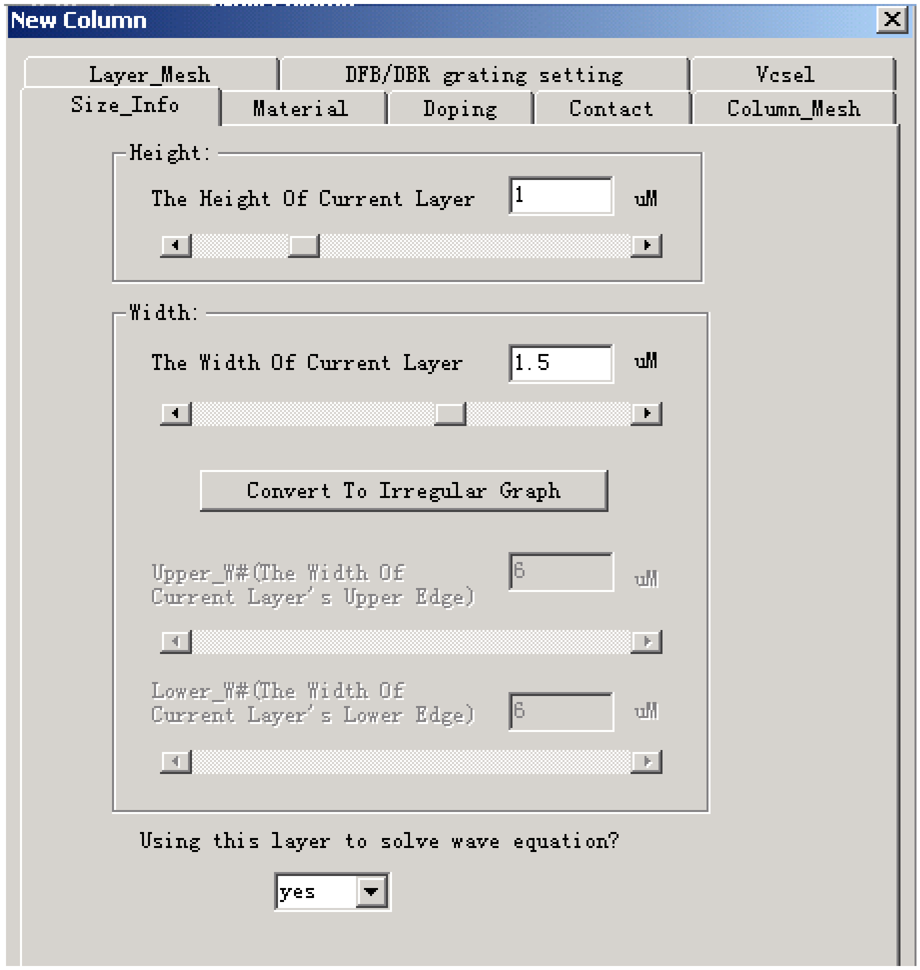


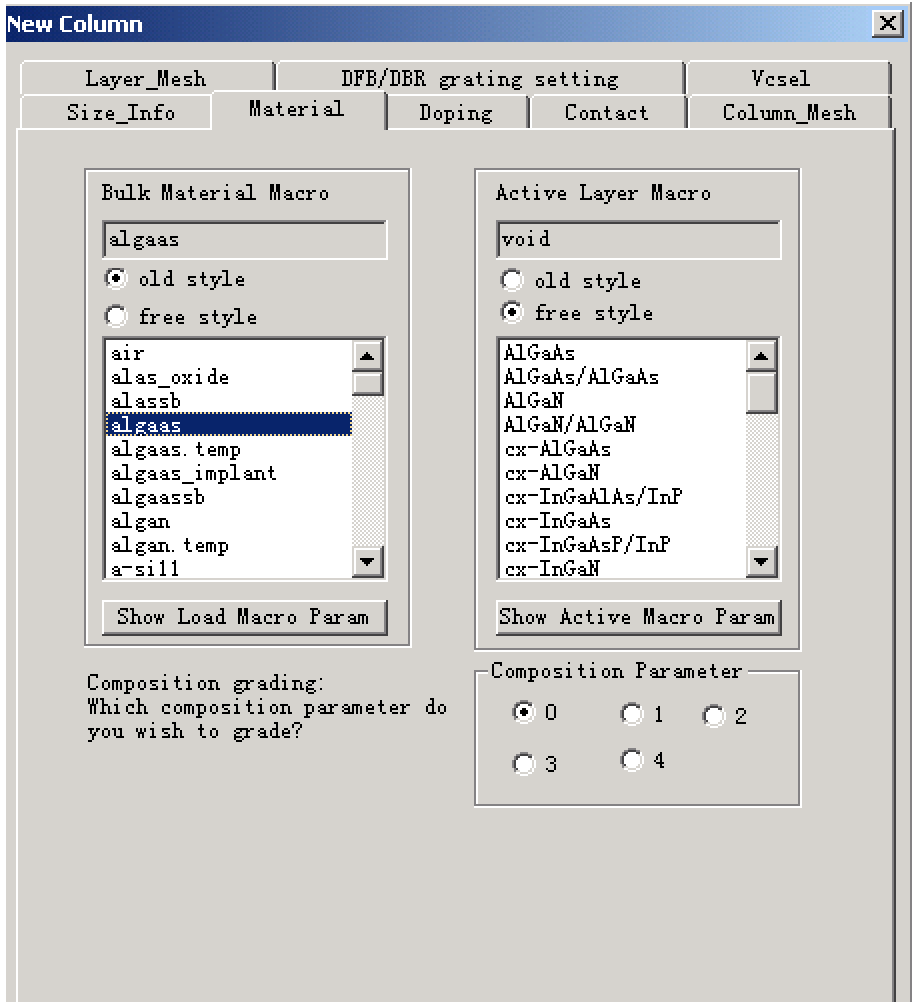
Figure 3: Property of New Column

3) Input size information of this layer in the 'New Column' dialogue box



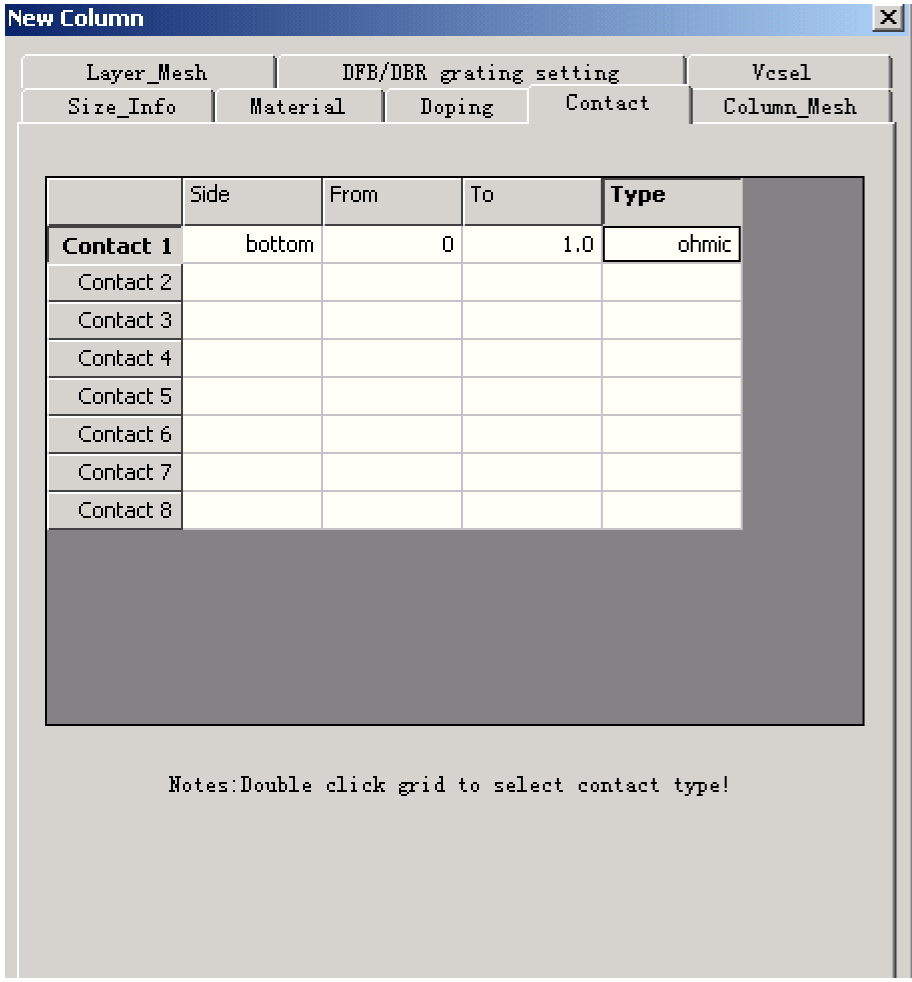
- Height: → 1
- Width: → 1.5

4) Material information of the first layer



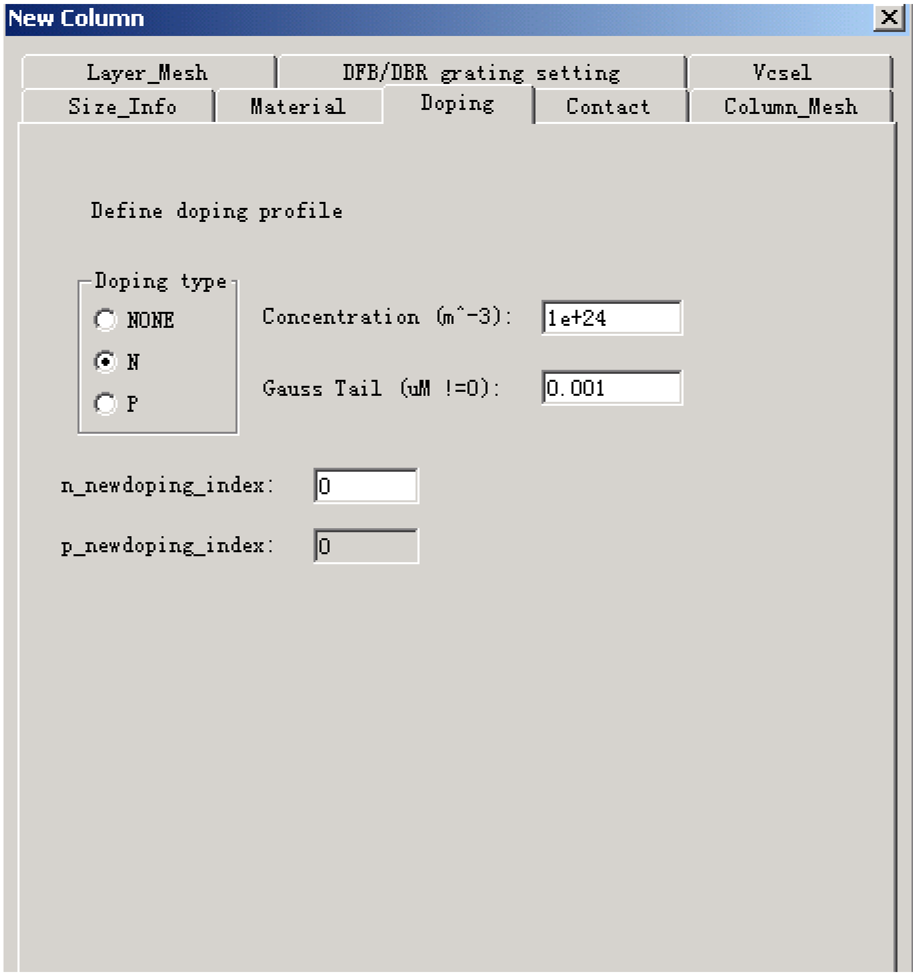
- Bulk material Macro → algaas
- Show bulk material param →0.5
- Active material macro → void

5) Contact information of the first layer



Side→ bottom
From → 0
To → 1.5
Type → ohmic

6) Doping information of this layer.



Doping type → n
Concentration → 1.e+24
Gauss Tail → 0.001

We have just finished entering the first layer of the column (layer1). Click OK to confirm.

At this point, the mesh has not yet been specified. We will define the mesh later using the 'auto mesh' function.

You should be able to see the 1st layer on the drawing area of the Layerbuilder (Refer to Figure 4). Keep in mind that you can always change your input by using Layer Edit →modify, or right-clicking the layer.

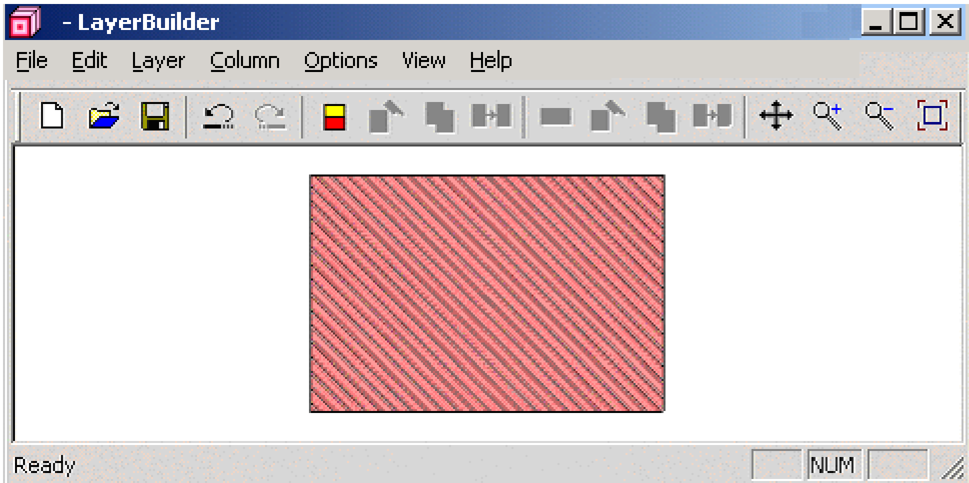


Figure 4 first layer finished

2. The second layer of the device

To add second layer (layer2), using your mouse, click on the layer you have just built and from LayerEdit→ NewLayer, the same layer dialog box as before will pop up. The input procedure is almost the same as the first Layer.

1) Size Information
    - Height: → 0.2
    - Width: → 1.5
2) Material Information
   This is an active layer and a little different from the first layer.
   I. Define Bulk material macro
      - Bulk material Macro → gaalas
      - Show bulk material param →0.1
   II. Define active layer
      - Active material macro → AlGaAs/AlGaAs
      - Xw → 0.1
      - Xb → 0.5
      - Temp → 300
3) Doping Information
   Doping type → none

3. The top layer of the only column

This layer is almost the same as layer1. We can perform the following trick by copying layer1 to layer3. We can accomplish this using the following steps:

- Click layer1 *OR* Layer Edit→copy.

- Click layer2 (since we need to copy layer1 just above layer2). Click Layer Edit → paste.

A new layer3 is created with exactly the same properties as layer1. We can then edit the values of layer3 by right clicking on layer3. The following procedure shows what we need to change from the first layer.

- Change: Doping → Doping type → p-type (from n-type)
- Add: Contact → Side → top
From → 0
To → 1.5
Type → Ohmic

Done! Your geometry input file has been completely finished now. Don't forget to save it!

Appendix:
Since we've skipped setting up column mesh/layer mesh allocation so far, it is necessary to have a brief look at it.
Column Mesh: Refer to Figure 5.
Layer Mesh: Refer to Figure 6.

The parameters for column and layer mesh allocation are almost the same.
Mesh number(Layer): number of mesh in an integer. Only the layers located at the first column (in the left) have this property. The layers parallel to them have the same mesh numbers as theirs.

Mesh number(Column): number of mesh for the column in integer. Only the bottom layer of each column has this property.
Mesh ratio: please refer to Figure 7.
Shift centre: shift from the centre of the layer/column.

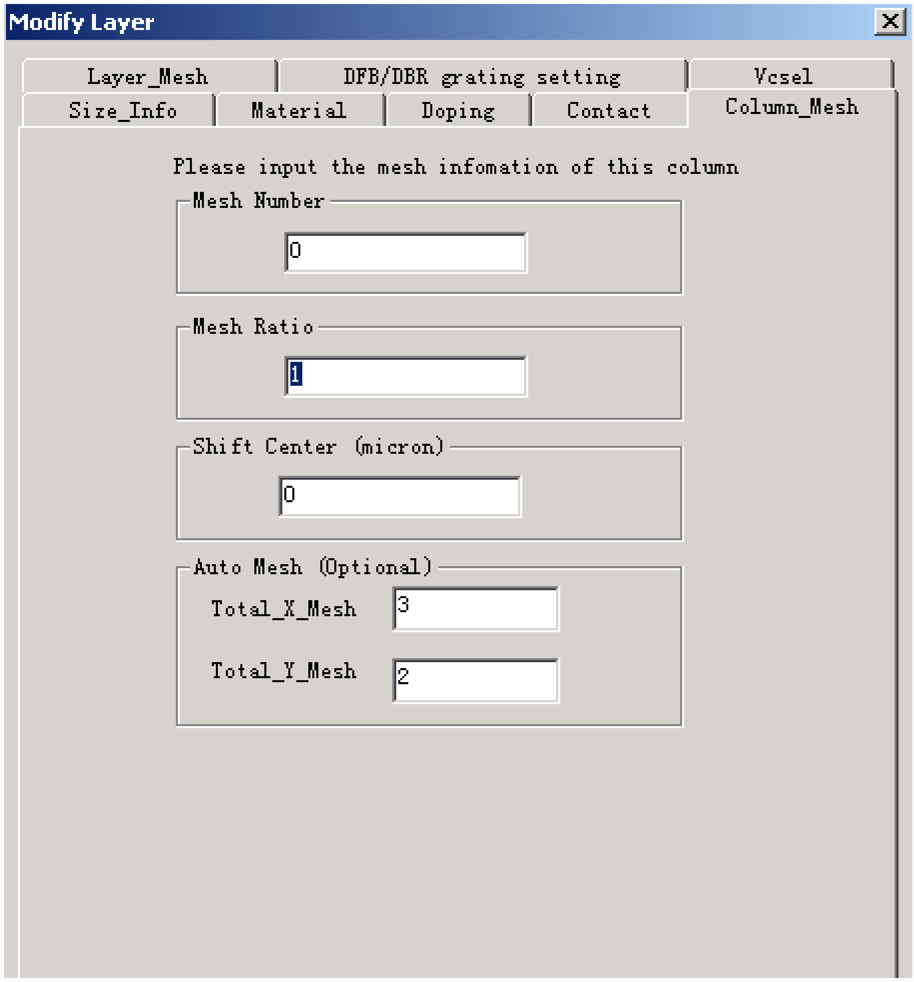


Figure 5 – Column Mesh

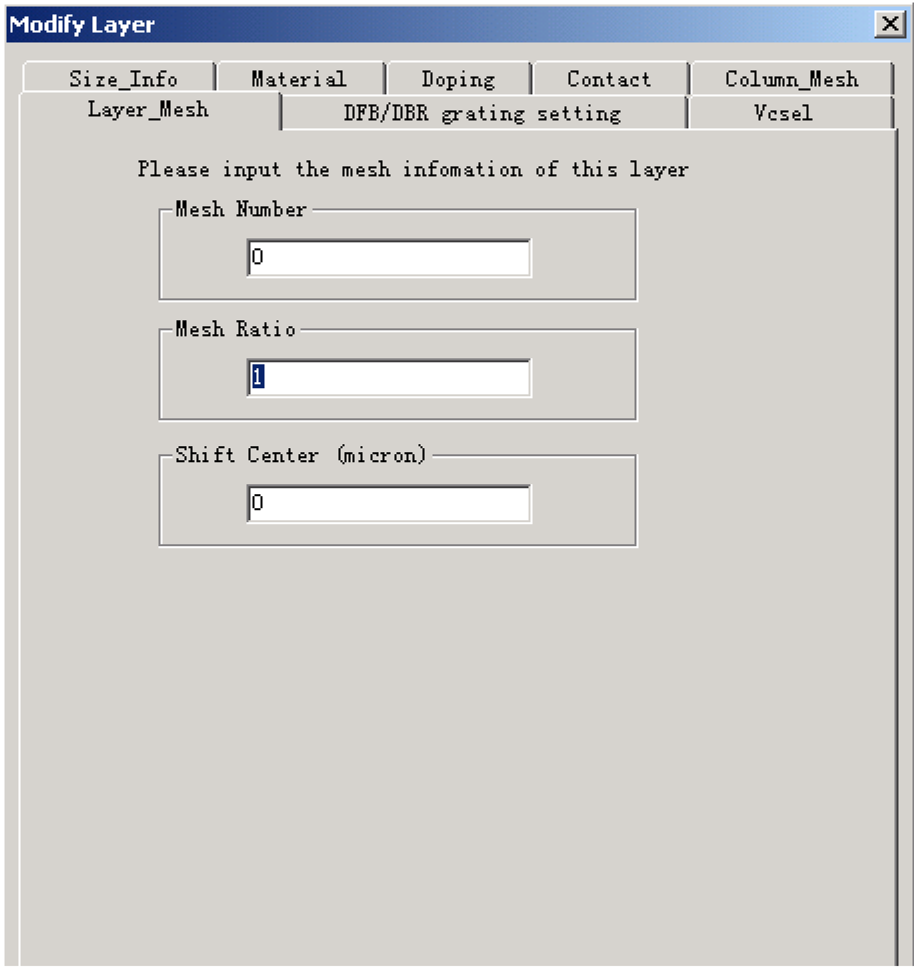
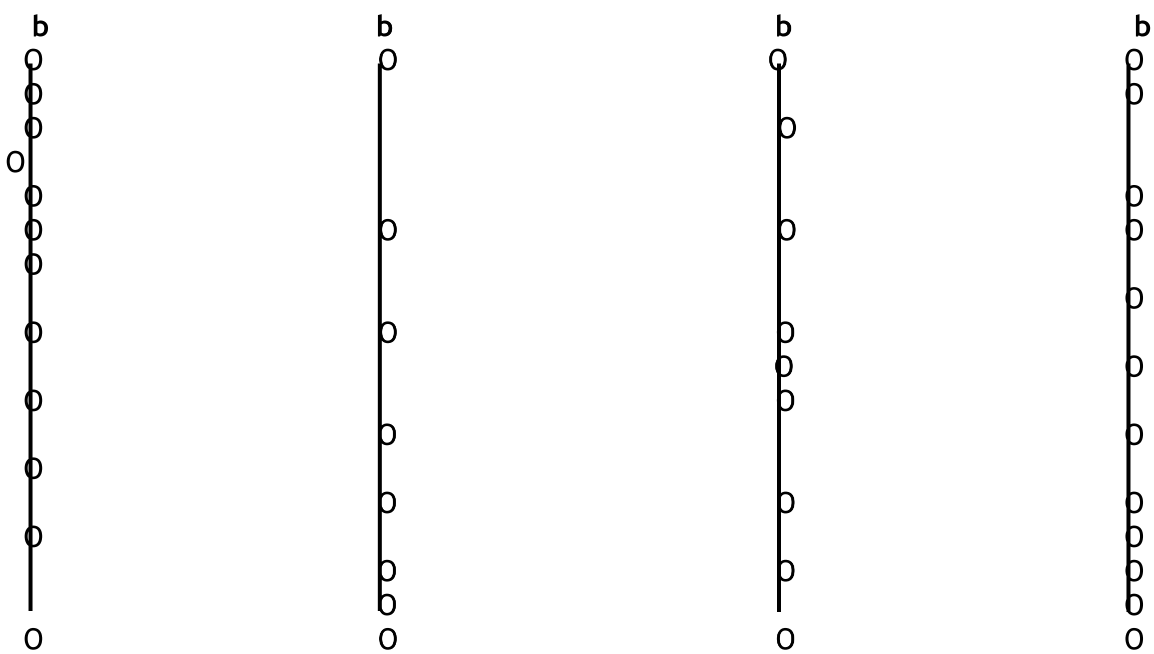


Figure 6 – Layer Mesh



a a a a

Ratio=0.8 Ratio=1.2 Ratio=0.8 Ratio=1.2

Figure 7: Demonstration of Mesh Ratio

3.2 Example 2

A 1D 5-layer GaAs/AlGaAs laser diode structure with the following schematics:
The original example is under *crosslig\examples\A_Layered_LD*

```
                    Top Contact
-----------------------------------------------------
           Al(0.71)Ga(0.29)As  (p-doped)
-----------------------------------------------------
            Al(x)Ga(1-x)As x=0.33->0.71
-----------------------------------------------------
                 GaAs active layer
-----------------------------------------------------
            Al(x)Ga(1-x)As x=0.71->0.33
-----------------------------------------------------
           Al(0.71)Ga(0.29)As ( n-doped)
-----------------------------------------------------
                  Bottom Contact
```

1. Set up the first layer of the only column of the device.

   In Layer builder, a device is made of columns that contain several layers. In order to build a layer, the column should be set up at first.

   1) Two ways to start a new column:
      - Click the short-cut button for "New Column" on the toolbar.
      - From menu "Column Edit" select "New Column."

   Refer to
   Figure 2: Setup a New Column

   A dialogue box will pop up. This is where you input the first layer structure on this column. (Refer to Figure 3)

   2) Size Information of the first layer
      - Height → 1.35
      - Width →1.5

   3) Material Information of the first layer

- Bulk mater macro → algaas
- Show bulk parameter → 0.71

4) Doping Information of the first layer.
   - Doping type → n
   - Concentration → 1.e+24
   - Gauss tail → 0.001

5) Contact Information of the first layer
   - Side → bottom
   - From → 0
   - To → 1.5
   - Type → Ohmic

We have just finished entering the first layer of the column (layer1). Click OK to confirm.

You should be able to see the 1st layer on the drawing area of the Layer builder. Keep in mind that you can always change your input from Layer Edit →modify, or right-clicking the layer.

2. The second/grading layer

To add a second layer (layer2), using your mouse, click on the layer you just built and from Layer -> NewLayer, the same layer dialog box as before will pop up. Input the data for this layer step by step like the first layer.

1) Size Information of the second layer
   - Height → 0.1462
   - Width →1.5

2) Material Information of the second layer
The bulk material in this layer has composition grading. So, it's a bit different from the usual bulk material definition.
   - Bulk mater macro → algaas
   - Composition parameter → 1
     (Define how many independent element compositions will be graded.)
     - Direction → +y
     - Grading_from → 0.71
     - Grading_to → 0.33

3) Doping Information of the second layer.

- Doping type → none

3. The third/active layer of the device

This layer is an active layer. Click on layer2 and then click LayerEdit → NewLayer.

1) Size Information
   - Height: → 0.0076
   - Width: → 1.5
2) Material Information
   This is an active layer and a little different from other layers.
   I. Define Bulk material macro
      - Bulk material Macro → gaalas
      - Show bulk material param →0

   II. Define active layer
      - Active material macro → AlGaAs/AlGaAs
      - Xw → 0
      - Xb → 0.33
      - Temp → 300
3) Doping Information

   - Doping type → none

4. The fourth/grading layer

This Layer is almost the same as the second layer. We can copy layer2 to layer4. Refer to the method of copying in Example 1.

We can then edit the values of layer4 by right clicking on layer4. In this case, we need to change the grading composition parameters.

- Composition parameter → 1
  (Define how many independent element compositions will be graded.)
  - Direction → +y
  - Grading_from → 0.33
  - Grading_to → 0.71

5. The top layer

Create top layer 5 by copying layer1 (Follow the same operation when copying layer 2 to 4). Once layer5 is created, do the following:

Change: Doping → Doping type → p-type (from n-type)

- Add: Contact → Side → top
  From → 0
  To → 1.5
  Type → Ohmic

Clicking ‘OK’ to confirm input and the following geometry will be displayed.

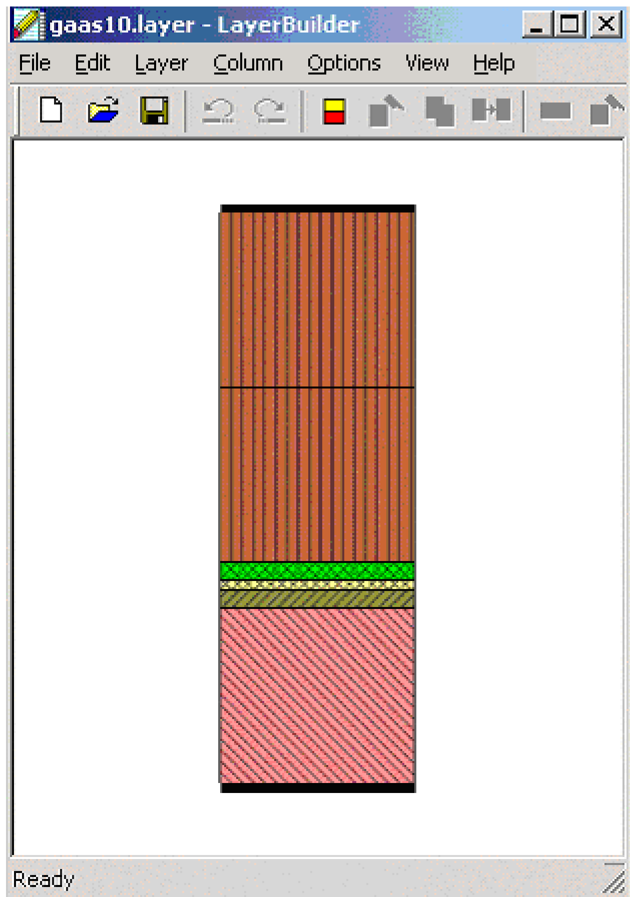


Figure 8: Geometry of gaas10

Done! Your .layer file has now been generated. Please don’t forget to save it. You can now proceed to the simulation process. Please refer to your manual for instructions on how to run a simulation.

3.3 Example 3

A 2D structure of the ridge-waveguide GRIN-SCH-SQW laser emits at 8.2 microns. The first column of this structure is similar to the previous example except that the top layer in the previous example is split into two layers.

Following steps (2) - (5) in the previous example, create 6 layers of structure with following properties:

layer1:
size_info = [0.8, 1.2], (in um)
material = bulkmaterial: ["algaas", 0.71], activemacro: ["void"]
composition parameter:["0"]
doping = ["n", 1e+024, 0.001]
contact = ["bottom", from=0, to=0.8], contact type: ["ohmic"]
layer2:
size_info = [0.8, 0.1462]
material = bulkmaterial: ["algaas", 0], activemacro: ["void"]
composition parameter: [1, "+y ", grade_from=0.71, grade_to=0.33]
doping = ["none", 0, 0.001]
contact = do not input anything, let it be default.
layer3:
size_info = [0.8, 0.076]
material = bulkmaterial["gaas"]
activemacro: ["AlGaAs/AlGaAs", 0, 0.33, 300]
composition parameter:[0]
doping = ["none", 0, 0.01]
contact = do not input anything, let it be default.
layer4:
size_info = [0.8, 0.1462]
material = bulkmaterial: ["algaas", 0], activemacro:["void"]
composition parameter:[1, "+y ", grade_from=0.33, grade_to=0.71],
doping = ["none", 0, 0.001],
contact = do not input anything, let it be default.
layer5:
size_info = [0.8, 0.0538]
material = bulkmaterial:["algaas", 0.71], activemacro:["void"]
composition parameter:[0]
doping = ["p", 1e+024, 0.001]
contact = do not input anything, let it be default.
layer6:
size_info = [0.8, 1.1]
material = bulkmaterial:["algaas", 0.71], activemacro:["void"]
composition parameter:[0]
doping = ["p", 1e+024, 0.001]
contact = ["top", from=0, to=0.8]

To create the 2D structure, repeat the above procedure and type in all the information. A quick way to accomplish the same task is to notice the second column structure is almost the same as the first column except the top layer (which is vacuum). Thus we can simply copy the first column and change the properties on layers on the second column. To do this first click any layer in the first column then click on Menu:

- ColumnEdit->copy, ColumnEdit->paste.

After the duplication, you can modify the width of the new column by changing the width of any layer of the new column. To do this, simply right click on any one of the layers in the new column and modify its width. Change the top layer of the second column by right clicking on the top layer and change the bulk material macro to "vacuum." Set the contact to "none".

Repeat step (10): assign "3" to "total_x_mesh," and "35" to "total_y_mesh." This will invoke "auto mesh" function. Then repeat Step(11) and save it as gaas20g. This will generate a .layer file acceptable to the Simulators.